



Option 4: Bank Statement Extraction Agent

Sources

Binder2_Redacted.pdf 53.8 MiB

ixonia_binder2_ocr.txt 468.6 KiB

Goal

Build an **Agent** that takes a bank statement and returns its **totals by periods** and **all transactions** as structured data.

One statement in → one structured result out.

Data Source

- `Binder2_Redacted.pdf` — original statement (Ixonía Bank, multiple periods, redacted descriptions)
- `ixonia_binder2_ocr.txt` — OCR of the same PDF

Use either or both. We will also evaluate on **unseen statements from other banks** — your solution must handle them **without code changes**.

What you build (3–6 hours)

A function:

```
def extract(pdf_path: str, txt_path: str | None = None) -> dict
```

returning:

```
{ "account": { "bank": "Ixonla Bank", "account_last4": "4664", "period":
{ "start": "2025-04-01", "end": "2025-04-30" } }, "summary": { "beginnin
g_balance": 597068.70, "ending_balance": 509121.59, "deposits_total": 12
14254.05, "deposits_count": 81, "withdrawals_total": 1302201.16, "withdr
awals_count": 111 }, "transactions": [ { "date": "2025-04-01", "descript
ion": "AIRLINEHYD 2759/VENDOR PMT", "deposit": 1809.28, "withdrawal": nu
ll } ] }
```

Plus a way to run it: CLI, HTTP endpoint, or one-field UI.

Requirements

- Values must come from the document. No hallucinated numbers.
- Must reconcile: $\text{beginning} + \Sigma \text{deposits} - \Sigma \text{withdrawals} = \text{ending}$. Flag mismatches explicitly.
- Generalize to new bank layouts via prompts / architecture — not code edits.

Grading

1. Summary fields — exact match.
2. Transactions — Must be reconciled with totals.
3. Generalization — same scoring on 2–3 unseen statements.

Deliverables

1. Repo link.
2. Short README: architecture, self-reported accuracy + known weaknesses.
3. 30-min demo: live run on the sample + one statement you haven't seen.

Etalon results for sample

Source: Binder2_Redacted.pdf

#	Period	Account	Deposits (count)	Deposits (\$)	Withdrawa
1	Apr 2025	1664	81	\$1,214,254.05	111
2	May 2025	1664*	95	\$926,416.11	142
3	Jun 2024	1664	63	\$1,050,851.95	99
4	Jul 2024	4664	84	\$848,578.92	82
5	Aug 2024	4664	83	\$1,178,227.39	88
6	Sep 2024	4664	71	\$1,085,703.81	118
7	Sep 2024	4623	13	\$336,565.07	35
8	Oct 2024	4664	83	\$1,187,061.65	96
9	Nov 2024	4664*	75	\$847,969.53	120
10	Dec 2024	4664	67	\$1,223,865.12	65